

Item

applicability

model boundary

SISO / MIMO

system order

properness

initial state

characteristic info

Conclusion

linear time-invariant systems only

maps input to output, not internal states

one transfer function is for one SISO channel

highest power of s in the denominator

for physical systems, denominator order $>$ numerator order

cannot represent nonzero initial-condition motion

$D(s)$ is the characteristic polynomial